

Fair Allocation, Prices and Quantum Sampling

A conceptual guide to *Fairness Is Advantage-Neutral*

Companion notes to the manuscript by Ritesh Roshan

Abstract

This guide explains the whole project from the beginning, without assuming a background in convex analysis, Markov chains or quantum computing. It follows one tiny example (three wireless links, two users) from the question “what is fair?” to the quantum circuit, and at each step says what the corresponding theorem in the manuscript does. Section 11 separates what is new from what was already known.

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1 The problem

Some resources cannot be split. A transmission slot goes to one link or another, a ventilator to one patient or another. When several people need them, we must decide who gets what, and we want the decision to be *fair* as well as efficient.

The running example. Three wireless links a, b, c sit in a row. Neighbouring links interfere, so a and b cannot transmit together, and neither can b and c . Links a and c are far apart and can. User 1 owns links a and c ; user 2 owns link b . Every link carries one unit of data when it transmits.

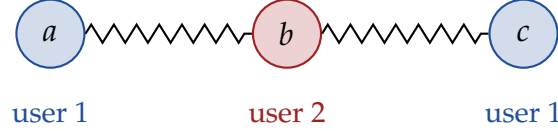


Figure 1: The conflict graph of the running example. A jagged edge means “cannot transmit at the same time”.

A *schedule* is a set of links that may transmit together: a set with no conflicting pair, which graph theory calls an *independent set*. There are five:

$$\emptyset, \{a\}, \{b\}, \{c\}, \{a, c\}.$$

The two interesting ones are $\{a, c\}$, which gives user 1 two units and user 2 nothing, and $\{b\}$, which gives user 1 nothing and user 2 one unit.

The difficulty in one line. Every single schedule leaves somebody with nothing. No deterministic choice is fair.

2 What does “fair” mean?

Economists have many answers, and the paper handles all of the standard ones [3, 23, 19, 18]. Write y_1, y_2 for what the two users receive.

Criterion	Formula	Plain meaning
Utilitarian	$y_1 + y_2$	maximise the total, ignore who gets it
Max-min (Rawls)	$\min(y_1, y_2)$	make the worst-off person as well off as possible
Nash welfare	$\log y_1 + \log y_2$	balance total and equality (proportional fairness)
α -fairness	$\sum_i y_i^{1-\alpha} / (1-\alpha)$	a dial from utilitarian ($\alpha = 0$) to max-min ($\alpha \rightarrow \infty$)
Gini-based welfare	$\text{mean} \times (1 - \text{Gini})$	penalise inequality in the whole distribution

All of these are *concave*: spreading a fixed amount more evenly never lowers them. That single property is all the theory needs. We use max-min as the running criterion because it is the easiest to picture.

2.1 Randomising: the lottery

Since every schedule is unfair, use a lottery: pick $\{a, c\}$ with probability p and $\{b\}$ otherwise. On average user 1 gets $2p$ and user 2 gets $1 - p$. These are equal when $p = \frac{1}{3}$, and then each user expects $\frac{2}{3}$.

Deterministic max-min value: 0. **Lottery max-min value:** $\frac{2}{3}$. This is *ex-ante* fairness: fair in expectation, before the coin is tossed. Weighted lotteries are already used to allocate scarce medicines [27, 21]. If the allocation is repeated every time slot, the long-run averages also approach $\frac{2}{3}$ each.

3 Why fairness is awkward for quantum optimisers

Quantum annealers and algorithms such as QAOA [8, 16] accept an objective that is a *quadratic* polynomial in yes/no variables $x_v \in \{0, 1\}$ (“does link v transmit?”). Many fairness criteria are not quadratic. The minimum of n yes/no variables equals their product $x_1 x_2 \cdots x_n$, which has degree n , and an absolute value $|u_1 - u_2|$, the building block of the Gini coefficient, already has degree three in a small example. Forcing such a criterion into quadratic form requires extra variables and penalty terms [5], which is why earlier quantum work on equity either used a quadratic stand-in for fairness or left fairness out of the Hamiltonian.

The paper’s first observation: this barrier exists only for single allocations. Once we allow lotteries, the barrier disappears, as the next section shows.

4 The key idea: fairness is a set of prices

4.1 The picture

Plot each schedule as a point (what user 1 gets, what user 2 gets): $\{a, c\} \mapsto (2, 0)$, $\{b\} \mapsto (0, 1)$, $\{a\}, \{c\} \mapsto (1, 0)$, $\emptyset \mapsto (0, 0)$. A lottery produces an *average*, and averages of points fill in the shaded triangle of Figure 2, the *convex hull*. The fairest lottery is the point of the triangle that is best for the chosen criterion; for max-min it is where the diagonal $y_1 = y_2$ leaves the triangle, at $(\frac{2}{3}, \frac{2}{3})$.

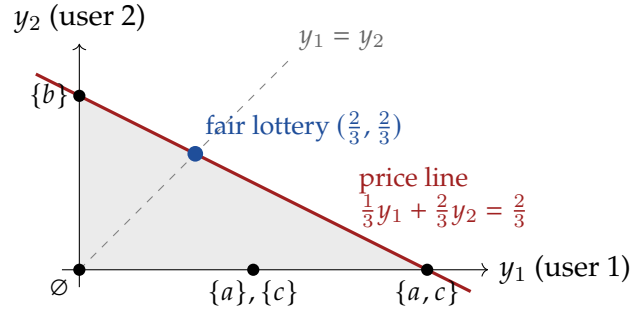


Figure 2: Every lottery lands in the shaded triangle. The fair point is where the diagonal leaves it. The red line, the “price line”, touches the triangle exactly there and passes through the two schedules that the fair lottery mixes.

4.2 Prices

Now give each user a *priority weight*, or price, $\theta_i \geq 0$ with $\theta_1 + \theta_2 = 1$, and ask a purely combinatorial question: *which schedule has the largest priority-weighted value $\theta_1 y_1 + \theta_2 y_2$?* At $\theta = (\frac{1}{3}, \frac{2}{3})$ the schedules $\{a, c\}$ and $\{b\}$ tie at $\frac{2}{3}$, which is exactly the fair value. That is not a coincidence:

Separation theorem (Theorem 2 of the paper). For any concave fairness criterion,

$$\underbrace{\max_{\text{lotteries}} \text{fairness}}_{\text{normative question}} = \min_{\text{prices } \theta} \left[\underbrace{\max_{\text{schedules } x} \theta \cdot u(x)}_{\text{combinatorial question}} - \underbrace{\Psi_*(\theta)}_{\text{penalty set by the criterion}} \right].$$

For max-min the penalty is zero and the prices range over all weights summing to one. The *fairness criterion* only says which prices are allowed. The *feasible schedules* only enter through the question “what is the best schedule at these prices?”.

The mathematics is classical convex duality, the same duality that turns a constrained problem into a Lagrangian with multipliers [24, 6]. The contribution is the reading: the normative part (what is fair) and the combinatorial part (what is feasible) separate cleanly.

The SVM analogy. A support vector machine finds a maximum-margin line, and at the optimum only the *support vectors*, the points touching the margin, carry nonzero dual weight. Here, at the optimum only the *worst-off users* carry nonzero price. In the example both users are worst-off together, so both prices are positive, and the price line touches the triangle at the two schedules the fair lottery mixes.

4.3 Two consequences

1. **No representation cost (Corollary 3).** The question “best schedule at prices θ ” is a *weighted* version of the usual problem, and it has the same polynomial degree as the utilities. In scheduling it is the maximum-weight independent set problem, which is quadratic and native to quantum hardware. Fairness never has to be written into the Hamiltonian at all.
2. **Advantage neutrality (Theorem 4).** Since fair allocation reduces to repeatedly solving the weighted problem, and the weighted problem is itself a special case of fair allocation, the two are equally hard on *any* computer, classical or quantum. Fairness therefore cannot be the source of a quantum speedup. If a quantum computer helps, it helps with the underlying combinatorial problem.

5 Doing it over time: priorities that learn

We usually do not know the fair prices in advance. The paper finds them by *learning* them while allocating, round after round, with the multiplicative-weights (Hedge) rule [9, 2]:

The update. After each round, multiply each user’s priority by $e^{-\eta \cdot \text{what they received}}$, then rescale so the priorities sum to one. Users who received a lot lose priority; users who received little gain it.

Here is the loop on the running example, with $\eta = 0.5$ and, for now, each round simply taking the best schedule at the current prices.

Round	Priorities (θ_1, θ_2)	Best schedule	Received	Running average	Worst-off average
1	(0.50, 0.50)	$\{a, c\}$	(2, 0)	(2.00, 0.00)	0.00
2	(0.27, 0.73)	$\{b\}$	(0, 1)	(1.00, 0.50)	0.50
3	(0.38, 0.62)	$\{a, c\}$	(2, 0)	(1.33, 0.33)	0.33
4	(0.18, 0.82)	$\{b\}$	(0, 1)	(1.00, 0.50)	0.50
5	(0.27, 0.73)	$\{b\}$	(0, 1)	(0.80, 0.60)	0.60
6	(0.38, 0.62)	$\{a, c\}$	(2, 0)	(1.00, 0.50)	0.50
7	(0.18, 0.82)	$\{b\}$	(0, 1)	(0.86, 0.57)	0.57
8	(0.27, 0.73)	$\{b\}$	(0, 1)	(0.75, 0.62)	0.62

From round 3 on, the priorities cycle around the fair price $\theta_1 = \frac{1}{3}$, and the schedules settle into the pattern $\{a, c\}, \{b\}, \{b\}$, which is exactly the fair lottery ($\frac{1}{3}$ and $\frac{2}{3}$). The worst-off average climbs toward $\frac{2}{3}$. Nobody told the algorithm the answer; it was discovered by the priorities.

The Grover analogy. Grover’s search repeats two simple moves: flip the sign of the target, then reflect every amplitude about the mean. The fairness loop repeats two simple moves: pick a schedule at the current prices, then lower the priority of users who did better than average and raise it for those who did worse. In both, one simple step is repeated, and the repetition does the work.

Regret guarantee (Theorem 9). After T rounds, the worst-off user’s average is at least the best achievable value minus an error that shrinks like $1/\sqrt{T}$, plus two small terms explained next.

6 From “best schedule” to “random good schedule”

Finding the single best schedule at given prices is itself a hard problem in large networks (maximum weight independent set is NP-hard). The paper therefore replaces “take the best” by “draw a random schedule that is *probably* good”, using the Gibbs, or Boltzmann, distribution from physics:

$$\Pr(\text{schedule } x) \propto e^{\beta \times \text{priority-weighted value of } x}.$$

The dial β is an inverse temperature. At $\beta = 0$ every schedule is equally likely; as $\beta \rightarrow \infty$ only the best one survives. At the fair price $\theta = (\frac{1}{3}, \frac{2}{3})$ with $\beta = 3$ the example gives:

Schedule	$\emptyset$	$\{a\}$	$\{c\}$	$\{b\}$	$\{a, c\}$
Probability	0.8%	5.9%	5.9%	43.7%	43.7%

The two tight schedules dominate, and weak ones still appear occasionally. The price of this softness is a small, known loss in fairness, the $\log |F|/(\beta n)$ term of Theorem 9 (Lemma 7); raising β shrinks it.

Why sample at all? Sampling is far easier than exact optimisation, and among all lotteries with a given average the Gibbs lottery is the most random, or maximum-entropy, one (Theorem 8). It commits to nothing beyond what fairness requires.

7 How to draw a random schedule: Markov chains

Listing all schedules is impossible in large networks, so the paper draws a sample with a random walk, the Metropolis rule [17, 10]:

1. From the current schedule y , **propose** a new schedule x , for example by switching one link on or off.
2. If x breaks an interference rule, stay at y .
3. If x is more valuable, **move** to x . If it is less valuable, move with probability $e^{\beta \times (\text{value}(x) - \text{value}(y))}$, otherwise stay.

Repeated long enough, the walk visits each schedule with exactly its Gibbs probability. The only requirement on the proposal is *symmetry*: proposing $y \rightarrow x$ must be exactly as likely as proposing $x \rightarrow y$.

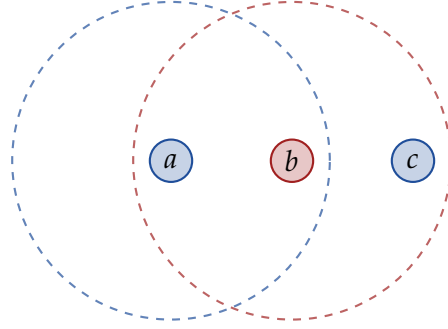
How long is “long enough”? The spectral gap. Think of shuffling a deck. Moving one card at a time takes thousands of moves to randomise it; a riffle shuffle takes about seven. The speed at which a random walk forgets where it started is measured by one number, the *spectral gap* γ : roughly $1/\gamma$ steps are needed [15]. Single-link flips are the “one card at a time” shuffle. They get stuck when good schedules are far apart, as $\{a, c\}$ and $\{b\}$ are: going from one to the other requires passing through worse schedules.

Warm-start theorem (Theorem 11). Priorities change only slightly per round, so the target distribution changes only slightly, and the walk never has to start from scratch. Each round needs about $\frac{1}{\gamma} \log \frac{1}{\epsilon}$ steps, *independent of the number of schedules*. Hence the cost of fair allocation is governed by one number, the spectral gap γ of the proposal, and improving the proposal is the only place where a quantum computer can help.

8 Where the quantum computer enters

8.1 Rydberg atoms already know the rules

In a Rydberg atom array, each atom is one link. An atom in its excited (Rydberg) state means “this link transmits”. Two excited atoms that are too close repel so strongly that the state cannot be reached: the *Rydberg blockade* [11, 4, 7]. If atoms are placed so that “too close” means “interfering”, the hardware physically cannot produce an infeasible schedule.



dashed circles = blockade radius of a (blue) and b (red):
 b lies inside a 's circle, c does not; b 's circle contains both a and c

Figure 3: The conflict graph of Figure 1 laid out as atoms. Blockade enforces the interference rules for free.

8.2 The quantum proposal

Instead of flipping one link, the proposal lets the atoms evolve for a short time under two effects: a *driver* (the PXP Hamiltonian [26]) that tries to switch each link on or off, but only if all its neighbours are off, and a *cost* term that makes valuable schedules gain phase. Measuring the atoms gives the proposed schedule. Three facts make it exactly correct:

1. **Never infeasible (Lemma 12).** Every step respects blockade, so the quantum state stays inside the set of valid schedules.
2. **Exactly symmetric (Lemma 14).** The evolution is built as a palindrome, running forwards then mirror-image backwards within each step, from real symmetric pieces. Such a product equals its own transpose, so “ $y \rightarrow x$ ” and “ $x \rightarrow y$ ” have exactly the same probability, at any circuit depth.
3. **Exact target (Theorem 15).** A symmetric proposal inside the Metropolis rule gives exactly the fair Gibbs lottery in the long run, whatever the quantum device does in between.

Intuition. The quantum evolution is a riffle shuffle of schedules that can jump between distant good schedules, such as $\{a, c\}$ and $\{b\}$, in one proposal, while the classical accept/reject coin keeps the answer exactly right. The quantum part only needs to propose well; it never has to be trusted to compute the answer.

The paper also gives the gate-level circuit, with exact gate counts, for devices without native blockade, and checks it against exact simulation to about 10^{-15} .

9 The whole algorithm on one page

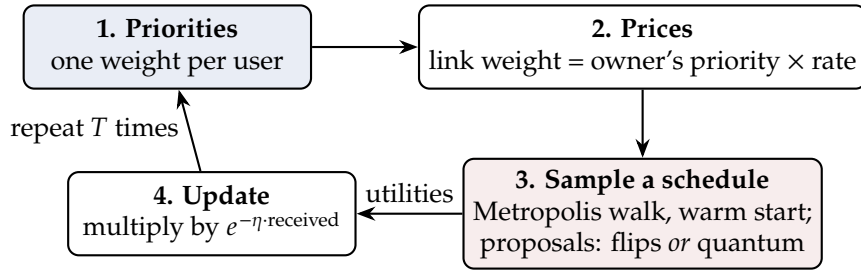


Figure 4: The fairness loop. Fairness lives in steps 1, 2 and 4; the quantum device lives only in step 3.

1. Start with equal priorities.
2. Turn priorities into link weights.
3. Take a few Metropolis steps from the previous schedule, mixing local flips with quantum proposals; the current state is this round's schedule.
4. Update priorities: whoever received less gains priority.
5. Repeat. The average allocation approaches the fairest lottery.

What this algorithm is called. There is no single established name, and the paper calls it guided quantum Gibbs dynamics. In standard terms the outer loop is *multiplicative weights* (Hedge), a no-regret, primal–dual method that solves the zero-sum game between “fairness” and “allocation” [9]. The inner loop is *Metropolis–Hastings MCMC* with a *quantum-enhanced proposal* [14]. The closest classical relative is adaptive CSMA in wireless networks [12].

10 What the experiments show

All numbers come from exact computation on networks of 8 to 20 links, in three families: unit-disk graphs (geometric, like real radio networks), Erdős–Rényi random graphs (dense) and random 3-regular graphs (sparse).

- **Faster mixing, always.** The quantum proposal increases the spectral gap over single flips in every instance, by factors of 3.4 to about 1100, with typical (geometric-mean) factors of 20 to 49 per family.
- **Better scaling, sometimes.** On dense random graphs the gap shrinks with network size about half as fast as for the best classical proposal. On unit-disk graphs this holds only at high temperature, and on sparse 3-regular graphs a classical swap move scales better.
- **Fewer steps per round.** Along real fairness trajectories, the quantum proposal needs the fewest steps in 82% of 39 instances: 21 steps on average, against 32 for an idealised uniform proposal and about 1900 for single flips.
- **Fairer outcomes at equal effort.** In a 1500-round max-min run on a 12-link network, the quantum-mixed chain reached a worst-off average of 1.327 ± 0.007 against 1.179 ± 0.139 classically, with the best possible value 1.460. It was also nineteen times more consistent from run to run.

Honest summary. These are large constant-factor gains and instance-dependent scaling gains at small sizes. They are *not* evidence of a general quantum advantage, and the paper says so. Known limits apply: quantum proposals cannot help on unstructured problems [20], classical solvers are strong on unit-disk graphs [1], and a quantum step costs more time than a classical flip.

11 What is new and what was already known

Already known	Where
Concave fairness criteria and their duals	welfare economics, convex analysis [3, 24]
Max-min fair lotteries via repeated weighted welfare	Kawase and Sumita [13]
Gibbs sampling of schedules with dual weights	adaptive CSMA [12]
Chains tracking slowly changing weights	network adiabatic theorem [22]
Quantum proposals inside Metropolis	quantum-enhanced MCMC [14]
Rydberg sampling of independent-set Gibbs states	Wild et al., by slow adiabatic preparation [28]
New in this project	
• The separation theorem for <i>every</i> concave criterion, and its consequence that fairness is advantage-neutral: it cannot create or destroy a quantum speedup. • An explicit warm-start bound whose cost does not grow with the number of schedules. • A quantum proposal that is feasible by physics and exactly symmetric by construction at any depth, given down to the gate level, used for the first time inside a fairness-driven sampler. • An open, verified implementation with exact benchmarks.	

12 Open questions

1. **One formula for the total cost.** Combining Theorems 9 and 11 should give “total steps $\approx$ (rounds needed for fairness) $\times$ (steps per round) $\approx \frac{\rho^2 \ln n}{\epsilon^2} \cdot \frac{1}{\gamma} \log \frac{1}{\epsilon}$ ”, making “the spectral gap is everything” a single theorem.
2. **When does quantum help?** Sampling independent sets is classically hard only above a sharp threshold of the weights, the uniqueness threshold [25]. Does the quantum gain switch on exactly there? A single sweep across the threshold would answer it.
3. **Noise.** Real hardware breaks the exact symmetry slightly. How much bias does that cause? A standard perturbation bound suggests about (asymmetry)/ γ .
4. **Scale and fair baselines.** Larger networks (30–40 links) with stronger classical competitors, such as parallel tempering and quantum-inspired proposals.

Glossary

Independent set. A set of links with no two in conflict; a valid schedule.

Lottery (ex-ante). A probability distribution over schedules; fairness judged on averages.

Concave criterion. A fairness score that never decreases when outcomes are spread more evenly.

Convex hull. All averages of a set of points; the set of outcomes lotteries can achieve.

Prices / dual weights θ . Priority weights on users; the fairness criterion decides which are allowed.

Weighted welfare. $\sum_i \theta_i y_i$; the only question the combinatorial part ever answers.

Multiplicative weights (Hedge). Update rule that raises the priority of users who received little.

Gibbs distribution, β . Lottery with probability $\propto e^{\beta \cdot \text{value}}$; β sets how greedy it is.

Metropolis rule. Propose a move, accept with probability $\min\{1, e^{\beta \Delta \text{value}}\}$.

Spectral gap γ . Mixing speed of a random walk; about $1/\gamma$ steps to forget the start.

Rydberg blockade. Nearby excited atoms are forbidden, so the hardware enforces conflicts.

PXP driver. Switches an atom only if all its neighbours are off.

Palindromic Trotter step. A product of gates that reads the same forwards and backwards; it makes the proposal exactly symmetric.

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